

B6 Transmission Headroom and Scottish Wind Curtailment: A Day-Ahead Predictive Analysis of the GB Electricity System

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Abstract

This project tests whether published day-ahead transmission information improves prediction of subsequent Scottish wind curtailment activity in the Balancing Mechanism. The analysis reconstructs B6/SCOTEX boundary conditions from NESO data and aligns them with Scottish wind Bid-Offer Acceptances, day-ahead wind forecasts and day-ahead demand forecasts. The broader descriptive sample contains 42,332 settlement periods from 22 April 2024 to 30 September 2026, while the main predictive sample contains 31,532 periods for which wind-forecast publication timing can be verified.

Lower B6 headroom is strongly associated with both a higher probability and a larger average volume of downward Scottish wind BOA activity. After controlling for forecast wind, forecast demand and calendar effects, adding B6 headroom raises event-model ROC AUC from 0.817 to 0.862 and average precision from 0.909 to 0.936. Its contribution to positive-event magnitude is smaller and nonlinear. In the complete two-part forecast, applying the same chronological probability-calibration procedure to both event models, the B6-enhanced specification reduces MAE from 899.8 MW to 860.1 MW and raises out-of-sample R^2 from 0.47 to 0.50. Robustness checks show that the result is not driven by the unusual 10,000 MW B6-limit state or by a single definition of B6 stress. The findings are predictive rather than causal.

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1 Introduction

A large share of Britain’s wind generation is located in Scotland, while a significant share of electricity demand is located further south. As renewable generation increases, greater volumes of electricity may need to move from areas of high generation to areas of higher demand. These transfers are constrained by the physical capability of the transmission network.

This project focuses on the B6 transmission boundary, which separates the SP Transmission and National Grid Electricity Transmission networks and forms an important interface for electricity transfers from Scotland into England. During periods of high north-to-south transfer, forecast flows can approach the published transfer capability of the boundary.

When transmission capability becomes scarce, the National Energy System Operator (NESO) may need to take balancing actions to maintain secure operation of the electricity system. These actions can include reducing generation on one side of a constraint and adjusting generation, demand or other resources elsewhere. For Scottish wind generators, this can result in downward Balancing Mechanism actions during periods when available generation cannot be accommodated by the wider system.

The purpose of this project is to examine whether information available at the day-ahead stage helps identify when this type of balancing activity is more likely to occur. The analysis focuses on the relationship between forecast pressure on the B6 boundary and subsequent Scottish downward wind Balancing Mechanism activity.

A central measure used throughout the analysis is B6 transmission headroom. For settlement period t , headroom is defined as

$$H_t^{B6} = L_t^{B6} - F_t^{B6}, \quad (1)$$

where L_t^{B6} is the published day-ahead transfer limit for the B6 boundary and F_t^{B6} is the corresponding forecast flow in the published positive direction. Lower headroom therefore indicates that forecast flows are closer to the available transfer capability, while negative headroom indicates that forecast flow exceeds the published limit in that direction.

The analysis does not assume that B6 is the sole driver of Scottish downward wind actions. The Scottish transmission system contains several interconnected and nested constraint boundaries, while balancing actions may also reflect wider network and system conditions. B6 is therefore treated as a case study for testing whether published transmission information contains useful predictive information about subsequent balancing outcomes.

The project first reconstructs historical day-ahead B6 conditions and aligns them with Scottish wind Balancing Mechanism actions. It then examines the descriptive relationship between B6 headroom and downward wind activity, before testing whether

B6 headroom adds predictive information beyond forecast wind, forecast demand and calendar conditions. A two-stage modelling framework is used to distinguish between the probability that downward wind activity occurs and the magnitude of activity when it does occur.

2 Motivation

Increasing renewable generation does not remove the physical limits of the electricity network. Strong wind output can increase the amount of low-cost generation available while also increasing pressure on transmission infrastructure. Where transfer capability is insufficient, NESO may need to change generation or demand through the Balancing Mechanism to keep the system secure.

The timing of information therefore matters. Explaining a balancing action after delivery is useful for understanding the market, but less useful for decisions that must be made beforehand. This project treats forecasting as a decision-support problem: whether information published at the day-ahead stage can distinguish relatively low-risk periods from periods in which material Scottish downward wind action is more likely.

B6 provides a useful case study because it is an observable Scotland-to-England transmission interface and because its published day-ahead flows and limits can be compared with subsequent Balancing Mechanism outcomes. The aim is not to reproduce NESO's full network model, but to test whether one transparent measure of transmission conditions adds information beyond more general wind and demand conditions.

3 Research Questions

The project is organised around three questions:

1. What relationship exists between day-ahead B6 transmission headroom and subsequent Scottish downward wind Balancing Mechanism activity?
2. Does B6 headroom provide additional predictive information once forecast wind, forecast demand and calendar conditions are taken into account?
3. How accurately can periods associated with elevated Scottish downward wind Balancing Mechanism activity be identified using information available at the day-ahead stage?

4 GB Electricity System and the B6 Boundary

4.1 Transmission Constraints in the GB Electricity System

Electricity generation and demand must be balanced continuously, but the transmission network also has to remain within its physical operating limits. A transmission constraint occurs when the network cannot accommodate the pattern of power transfers that would otherwise result from generation and demand across the system.

These constraints are geographical as well as system-wide. Generation may be available in one part of Great Britain while the transmission network has insufficient capability to move all of that electricity to demand elsewhere. In these circumstances, the National Energy System Operator (NESO) may need to change the pattern of generation or demand in order to maintain a secure and feasible system.

Transmission boundaries provide a useful way of representing these limits. Rather than describing a single transmission line, a boundary groups together circuits that form an interface between different parts of the network. The amount of power that can be transferred across a boundary depends on the capability of the underlying network and the system conditions at the time.

4.2 Scottish Generation and North–South Power Transfers

The location of Scottish renewable generation makes north–south transmission capability particularly important. Scotland contains substantial onshore and offshore wind capacity, while a large share of GB electricity demand lies further south.

NESO identifies additional renewable generation, particularly onshore and offshore wind, as a major driver of increasing transfer requirements from Scotland towards England. Under the future system pathways considered by NESO, power flows across the Scottish transmission boundaries are generally expected to increase as additional generation connects in Scotland [1].

The direction of transfer is not fixed. During periods of lower Scottish generation, electricity can flow northwards to meet Scottish demand. However, the dominant challenge relevant to this project is the movement of large volumes of power southwards during periods of high Scottish generation.

As wind output changes, the required transfer across the network can also change considerably. Transmission capability therefore becomes particularly important during periods in which high renewable output in Scotland coincides with insufficient local demand or insufficient capability to export that electricity further south.

4.3 The B6 Transmission Boundary

B6 is the transmission boundary between the SP Transmission network in Scotland and the National Grid Electricity Transmission network in England. It is therefore an important interface for power transfers between Scotland and England.

NESO’s Electricity Ten Year Statement describes increasing transfer requirements across B6 as additional Scottish generation connects to the system, particularly onshore and offshore wind. The same analysis notes that the range of potential B6 flows is wide because wind generation is variable [1].

For planning purposes, NESO currently describes the base capability of B6 as approximately 6.7 GW, with the Harker–Moffat 400 kV circuit identified as the limiting thermal constraint [1]. This planning capability should not be interpreted as a fixed operational limit for every settlement period. The day-ahead constraint dataset used in this project contains published B6 limits that vary through time as expected network conditions change [2].

This distinction is important for the empirical analysis. The project does not attempt to infer constraint pressure from a single assumed B6 capacity. Instead, the published limit for each settlement period is compared with the corresponding published forecast flow. B6 headroom therefore reflects the information available for that specific period rather than a fixed engineering threshold.



Figure 1: Southern Scotland and the SCOTEX transmission boundary in NESO’s Scottish network diagram. The SCOTEX boundary is the B6 constraint group analysed in this project. The crop is reproduced from NESO’s published Scotland day-ahead limit-boundaries network diagram [6].

4.4 The Balancing Mechanism and Scottish Wind Actions

The Balancing Mechanism is one of the tools available to NESO for adjusting generation and demand close to real time. Participants submit bids and offers that allow their expected position to be changed. Accepted offers increase the level of energy on the system, while accepted bids reduce it [7].

For a wind Balancing Mechanism Unit, an accepted downward action can therefore represent a reduction in output relative to the level that would otherwise have been produced. Wind units frequently contribute to accepted bid volume, particularly during periods when network conditions require generation to be reduced.

This provides the link between transmission conditions and the outcome studied in this project. If high Scottish wind generation creates large north–south transfer requirements while available network capability is limited, NESO may need to reduce generation behind the relevant constraint and take corresponding actions elsewhere on the system.

The empirical analysis therefore uses downward Balancing Mechanism actions from identified Scottish wind units as its observed outcome. NESO describes the BOA volume field in its wind dataset as the volume of generation curtailed by the BOA, reported in MW [3]. The report nevertheless refers to the constructed outcome as Scottish downward wind BOA activity when discussing attribution, because the dataset does not identify B6 as the cause of each individual action.

5 Data

5.1 NESO Day-Ahead Constraint Flows and Limits

The primary transmission dataset contains day-ahead constraint information published by NESO [2]. It records forecast power flows and corresponding transfer limits for a range of transmission boundaries across Great Britain.

The raw dataset contains 728,529 observations across 31 constraint groups. The analysis isolates observations associated with the SCOTEX/B6 boundary. This produces 64,426 raw B6 records before duplicate handling and subsequent validation.

For each settlement period, the dataset provides the two quantities required to construct the main transmission variable used in the project: the published B6 transfer limit and the corresponding forecast flow.

NESO states that from 22 April 2024 its published day-ahead flows no longer include expected constraint mitigation or optimisation actions [2]. Because the statistical properties of B6 headroom differ substantially before and after this date, the primary empirical analysis is restricted to the post-change period rather than treating both regimes as directly comparable.

5.2 Scottish Wind Balancing Mechanism Actions

NESO’s Wind BOA Volumes dataset is used to measure subsequent downward activity from Scottish wind generators [3].

The source files contain 664,485 individual balancing actions. Each observation represents an action-level record rather than an aggregated settlement-period total. Multiple accepted actions may therefore occur for the same generating unit and settlement period.

Scottish wind units are identified using a separate mapping of Balancing Mechanism Unit identifiers. The mapping process resolves all 198 candidate identifiers, with 143 identified as Scottish generator IDs used in the wind-action analysis.

For each accepted action, downward volume is represented as

$$C_i = \max(-V_i, 0), \quad (2)$$

where V_i is the reported BOA volume for action i . Positive BOA observations are not netted against downward actions.

Action-level downward volumes are subsequently aggregated to settlement-period level across the identified Scottish wind units. The resulting Scottish wind dataset contains 26,236 settlement periods with observed downward BOA activity.

5.3 Day-Ahead Wind and Demand Forecasts

Day-ahead wind and demand forecasts are included as additional explanatory variables. The wind series comes from NESO’s *Historic Day Ahead Wind Forecasts* dataset [4], while the demand series comes from the *Day Ahead Half Hourly Demand Forecast Performance* dataset [5].

Forecast wind is particularly important because it is related to both the transmission conditions of interest and the subsequent balancing outcome. Higher forecast wind may increase north–south transfer requirements while also increasing the amount of wind generation potentially exposed to downward balancing action.

Publication timing is treated explicitly. Wind observations enter the verified-wind modelling sample only for dates where the historic forecast timestamp can be shown to precede delivery and where the settlement-day structure is complete, including valid 46- and 50-period clock-change days. The demand data include a publication timestamp and are aligned to the corresponding delivery settlement period.

Demand forecasts provide a broader measure of system conditions. Higher demand may absorb a larger share of available generation, while lower demand can increase the amount of power that must be transferred between regions.

These variables are treated as observable day-ahead controls rather than outcomes. Their inclusion allows the analysis to test whether B6 headroom provides predictive

information beyond more general system conditions.

5.4 Data Grain and Time Alignment

The final analytical grain is the GB settlement period.

B6 flow and limit observations, wind and demand forecasts, and Scottish wind balancing actions are aligned by settlement date and settlement period. Balancing actions are first aggregated from action level before being joined to the transmission and forecast data.

Settlement-period alignment requires particular care around daylight-saving transitions, when GB electricity-market days may contain 46 or 50 settlement periods rather than the usual 48. These observations are retained where they can be aligned unambiguously rather than forcing every calendar day into an assumed 48-period structure.

5.5 Sample Construction

Two related samples are used in the analysis.

The broader B6 and Scottish wind BOA dataset contains 42,332 settlement periods across 882 days between 22 April 2024 and 30 September 2026. This sample is used for descriptive analysis of B6 conditions and balancing activity.

The predictive modelling requires verified day-ahead wind forecasts in addition to B6, demand and BOA information. The resulting modelling sample contains 31,532 settlement periods across 657 days, covering 22 April 2024 to 23 April 2026.

Table 1 summarises the two principal analytical samples.

Table 1: Principal analytical samples

Sample	Periods	Days	Use
B6 and Scottish wind BOA	42,332	882	Descriptive analysis
Verified-wind modelling sample	31,532	657	Predictive modelling

The smaller modelling sample reflects the availability of verified wind-forecast data rather than an additional outcome-based filter. The distinction is maintained throughout the project so that descriptive results based on the broader sample are not inadvertently presented as if they came from the modelling sample.

6 Data Engineering and Validation

6.1 Raw Data Preservation and Ingestion

The data pipeline was designed to preserve the original source files while carrying out cleaning and transformation programmatically. Raw NESO files were therefore retained separately from the analysis-ready tables used later in the project.

The workflow uses Python for cleaning and type normalisation and PostgreSQL for structured storage, validation and construction of analytical tables. Separating raw, cleaned and modelling layers makes the transformations reproducible and reduces the risk of manual changes being introduced during analysis.

The final database distinguishes between raw source data, validated core tables, audit outputs and modelling marts. This also allows unusual observations to be investigated without modifying the original records.

6.2 B6 Boundary Identification and Methodology Change

The day-ahead constraint dataset contains 31 published constraint groups. B6 observations were identified using the SCOTEX constraint group, producing 64,426 raw records.

Initial validation identified 96 excess exact duplicates. These were removed while preserving four repeated displayed timestamps associated with the autumn daylight-saving transition, where the underlying observations contained different values and therefore represented legitimate settlement periods rather than duplicate records.

After duplicate handling, 64,330 B6 observations remained.

A more important issue was a structural change in the published data. B6 headroom before and after 22 April 2024 displayed substantially different distributions. Before the change, mean headroom was approximately 3,264 MW and no observations were negative. After the change, mean headroom fell to approximately 705 MW and around one third of observations were negative.

Because this change reflects a break in the published day-ahead flow methodology rather than a gradual change in market conditions, the primary analysis begins on 22 April 2024. Pre-change observations are retained for validation but are not pooled with the main empirical sample.

6.3 Settlement Period Alignment

All datasets are aligned using settlement date and settlement period.

A normal GB electricity-market day contains 48 half-hour settlement periods, but daylight-saving transitions create shorter and longer days. The pipeline therefore does not assume that every date contains exactly 48 observations.

The source-coverage audit identifies 885 complete days, alongside a small number of dates affected by missing source records or incomplete values. Two days are entirely missing, five contain partial source coverage and one contains a value-level omission. After the full set of quality and alignment rules is applied, the analytical B6 mart contains observations from 882 days.

The modelling mart retains valid 46-period spring clock-change days and 50-period autumn clock-change days where observations can be aligned consistently. This avoids

introducing artificial gaps or duplicate removal based only on displayed clock time.

6.4 Construction of B6 Headroom

For each settlement period, B6 headroom is calculated as

$$H_t^{B6} = L_t^{B6} - F_t^{B6}, \quad (3)$$

where L_t^{B6} is the published transfer limit and F_t^{B6} is the corresponding forecast flow.

The resulting variable measures the difference between the published capability and the expected transfer across the boundary. Negative values indicate that the published forecast flow exceeds the corresponding published limit in the positive direction.

Of the validated post-methodology-change B6 observations, 42,595 contain usable headroom values. The primary settlement-period mart contains 42,332 observations after the remaining completeness and alignment rules are applied.

An additional utilisation measure was constructed as forecast flow divided by the published limit. Eleven observations contain a zero published B6 limit and therefore have undefined utilisation. These observations are retained because headroom remains well defined and utilisation is not used as the main transmission predictor.

6.5 Construction of Downward BOA Measures

The Balancing Mechanism source data are recorded at action level, so several accepted actions may occur for the same unit within one settlement period.

The analysis does not remove these records as duplicates. Instead, downward volume is calculated at action level and then aggregated across the identified Scottish wind units for each settlement period.

For action i ,

$$C_i = \max(-V_i, 0), \quad (4)$$

where V_i is the reported BOA volume. Positive actions are assigned zero in the downward measure rather than being netted against negative actions.

The settlement-period outcome is then

$$C_t = \sum_{i \in t} C_i. \quad (5)$$

This produces a measure of observed Scottish downward wind BOA activity for each settlement period.

The terminology is important. The variable is used as an empirical measure of downward wind balancing activity rather than as a complete engineering attribution of curtailment

to B6. The source data identify accepted balancing actions, but do not establish that every action was caused specifically by the B6 boundary.

6.6 Scottish Wind BMU Mapping

A separate validation step was used to determine which Balancing Mechanism Units should be classified as Scottish wind generation.

All 198 candidate BMU identifiers were resolved during the mapping process. Of these, 143 generator identifiers were classified as Scottish and retained for construction of the wind-action outcome.

Performing this mapping explicitly avoids relying on partial string matching or assuming that every wind-labelled unit belongs to the relevant geographical area.

6.7 Data Quality and Completeness Checks

Several additional checks were carried out before modelling.

One unusual feature of the B6 data is a published transfer limit of exactly 10,000 MW. This occurs in 927 settlement periods, representing approximately 2.2% of the full analytical sample. These periods also display different balancing behaviour from the remainder of the sample.

The observations are not automatically treated as erroneous. The underlying published values are retained and a separate indicator variable is constructed for the 10,000 MW state. The main models therefore preserve the observations, while later robustness tests examine whether the results survive when they are excluded from both evaluation and model estimation.

The final B6 and Scottish wind BOA mart contains 42,332 settlement periods across 882 days. Only 386 settlement periods containing observed Scottish downward BOA activity are lost through the strict B6 quality and settlement-period alignment rules, corresponding to approximately 1.35% of the aggregate downward BOA measure.

The purpose of these validation steps is not to create a perfectly regular dataset by removing unusual observations. Instead, the approach distinguishes between genuine data errors, known market-calendar features and unusual but potentially real operating states, with each treated separately in the analysis.

7 Exploratory Analysis

7.1 Distribution of B6 Headroom

Across the full post-methodology-change sample, B6 headroom averages approximately 695 MW, with a median of 1,016 MW. The distribution is wide and includes a substantial

number of negative observations.

Approximately 33.5% of settlement periods have negative B6 headroom. The 10th percentile is approximately -2,763 MW, while the 90th percentile is approximately 3,294 MW. The full observed range extends from around -8,225 MW to 18,673 MW.

This distribution indicates that B6 conditions vary considerably across settlement periods. Negative headroom is common enough to form a meaningful part of the sample rather than representing a small number of isolated observations.

The upper tail is partly influenced by unusual published B6-limit states, including the 10,000 MW observations identified during validation. These are retained in the descriptive analysis but examined separately in the robustness section.

7.2 Scottish Downward Wind BOA Activity

Scottish downward wind BOA activity occurs frequently within the B6 analysis sample.

Of the 42,332 settlement periods in the full analytical dataset, 25,836 contain positive downward wind BOA activity. This corresponds to an event rate of approximately 61.0%.

Across all settlement periods, the mean downward BOA measure is approximately 1,151 MW, while the median is approximately 322 MW. Conditional on positive activity, the mean rises to approximately 1,886 MW and the median to approximately 1,556 MW.

The positive-event distribution is moderately right-skewed. The 90th percentile is approximately 4,188 MW and the maximum observed settlement-period value is approximately 7,455 MW.

This motivates treating occurrence and magnitude separately in the later modelling. A large number of settlement periods contain no downward activity, while positive observations vary substantially in size.

7.3 Headroom and Balancing Activity

The descriptive relationship between B6 headroom and Scottish downward wind BOA is strong.

To examine the relationship without imposing a particular functional form, settlement periods are divided into deciles according to B6 headroom. The lowest-headroom decile records downward wind BOA activity in approximately 99.8% of settlement periods.

The event rate then generally falls as headroom increases. The second and third deciles record event rates of approximately 96.3% and 90.0%, while the corresponding rates in the eighth and ninth deciles fall to approximately 32.6% and 20.3%.

The mean downward BOA measure follows a similar pattern. It is approximately 3,638 MW in the lowest-headroom decile, compared with approximately 189 MW in the ninth decile.

The relationship is not perfectly monotonic. The highest-headroom decile displays a rebound in both event frequency and average downward BOA activity. Subsequent investigation shows that this part of the distribution contains a disproportionate number of unusual high-limit observations, including the 10,000 MW B6 state.

Figures 2 and 3 summarise the unconditional relationship between day-ahead B6 headroom and subsequent Scottish downward wind BOA activity. Lower-headroom groups are associated both with a higher probability of downward BOA activity and with larger mean settlement-period downward BOA volumes.

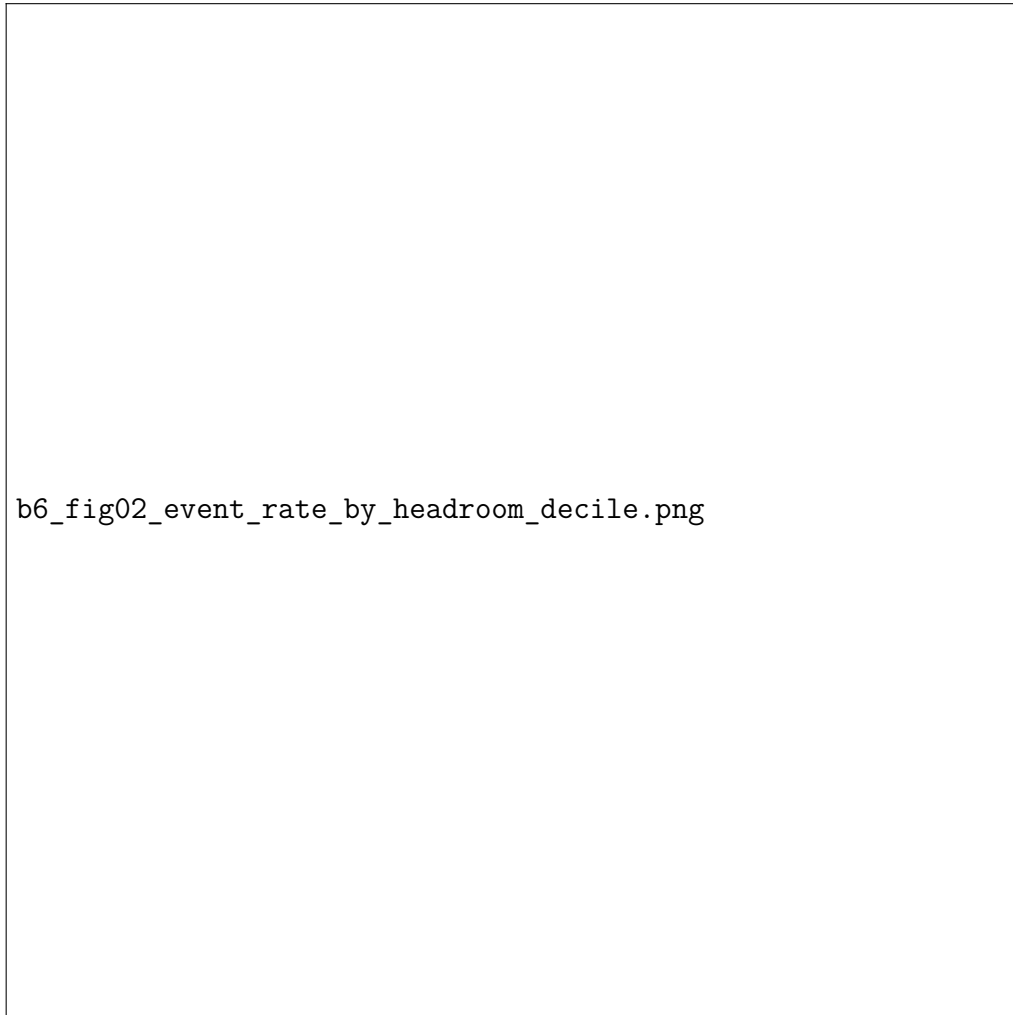


Figure 2: Scottish downward wind BOA event rate by day-ahead B6 headroom decile. Decile 1 contains the lowest B6 headroom values and decile 10 the highest. Event frequency declines strongly as headroom increases through most of the distribution, before rising in the highest-headroom decile. The figure is descriptive and does not establish that B6 caused the observed balancing actions.

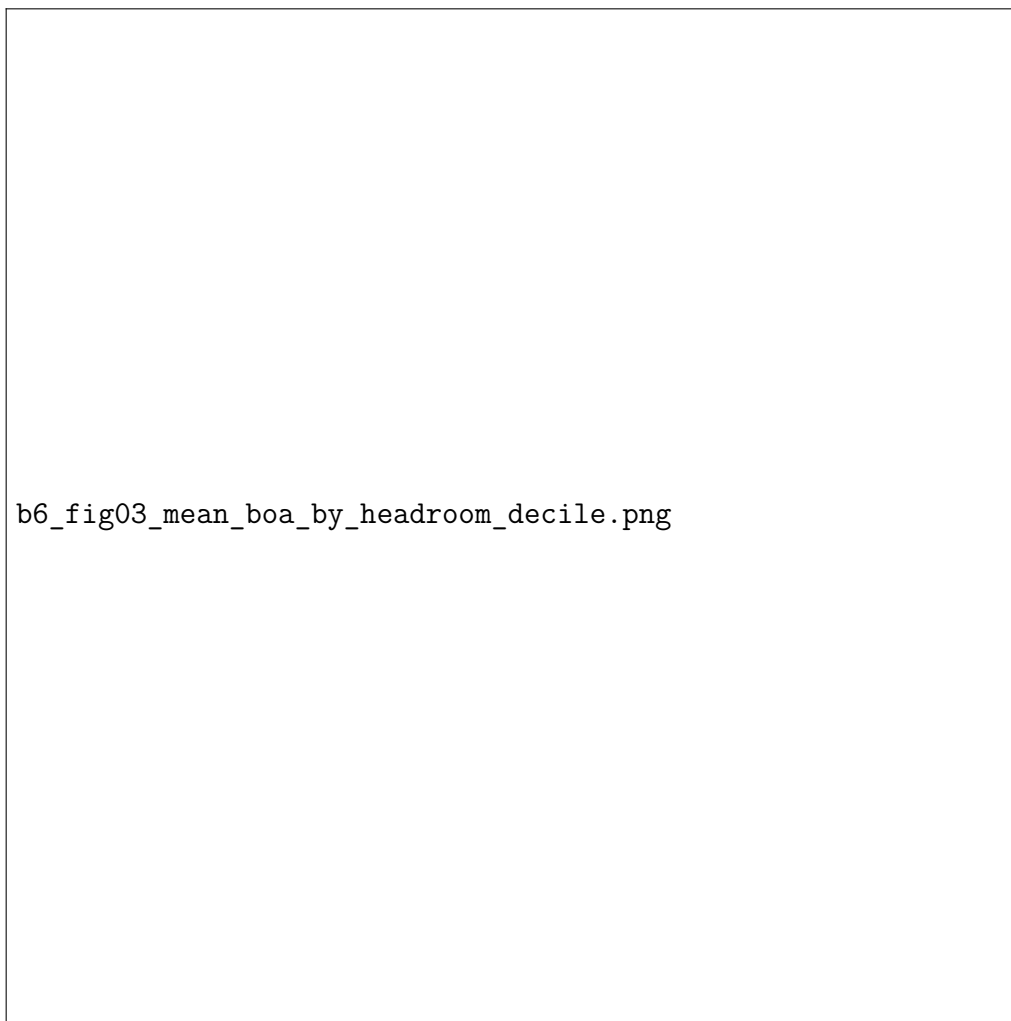


Figure 3: Mean Scottish downward wind BOA volume by day-ahead B6 headroom decile. Mean settlement-period downward BOA volume is highest in the lowest-headroom groups and generally declines as headroom increases, with a rebound in the highest-headroom decile.

7.4 The Role of Forecast Wind

The unconditional relationship between B6 headroom and downward BOA activity cannot be interpreted in isolation because forecast wind is related to both variables.

Within the verified-wind sample, forecast wind has a correlation of approximately -0.51 with B6 headroom. Higher forecast wind is therefore associated with tighter B6 conditions.

Forecast wind is also strongly associated with the subsequent balancing outcome. Its correlation with the downward BOA event indicator is approximately 0.58, while its correlation with downward BOA volume is approximately 0.74.

These relationships create an important confounding problem. A simple association between low B6 headroom and high downward BOA activity may partly reflect the fact that periods of high wind simultaneously increase north–south transfer requirements and increase the amount of wind generation available to be reduced.

To examine this more directly, the verified-wind sample is divided into forecast-wind quintiles and then further grouped by B6 headroom within each wind group.

Within the first four wind quintiles, the probability of downward BOA activity generally falls as B6 headroom increases. This indicates that B6 headroom contains information about event occurrence even when comparisons are made between settlement periods with broadly similar forecast-wind conditions.

The relationship becomes less informative in the highest forecast-wind quintile. At these extreme wind levels, downward BOA activity occurs in approximately 94–100% of settlement periods across the headroom groups. Event occurrence is therefore close to saturation, leaving comparatively little variation for B6 headroom to distinguish.

Figure 4 examines the same relationship within forecast-wind groups. The decline in event frequency with increasing B6 headroom remains visible across the first four wind quintiles. In the highest wind quintile, event frequency is already close to saturation across all headroom groups, leaving comparatively little variation for B6 headroom to distinguish.

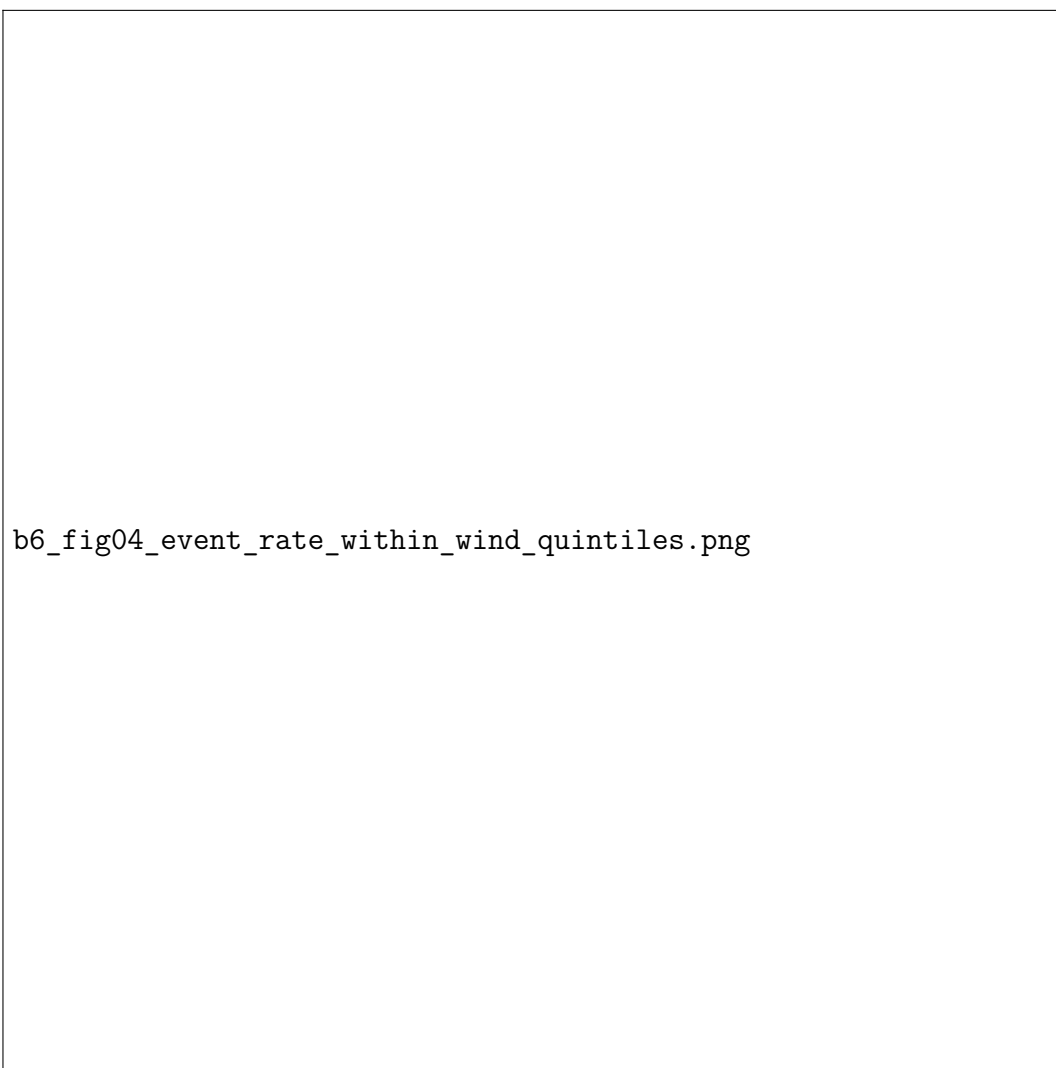


Figure 4: Scottish downward wind BOA event rate across B6 headroom groups within forecast-wind quintiles. Within the first four wind quintiles, event frequency generally falls as B6 headroom increases. In the highest wind quintile, event rates remain elevated across all headroom groups. This comparison is descriptive and should not be interpreted as a causal estimate of the effect of B6 headroom.

7.5 The 10,000 MW B6 Limit State

A published B6 limit of exactly 10,000 MW occurs in 927 settlement periods, representing approximately 2.2% of the full analytical sample.

These observations display noticeably different balancing behaviour from the rest of the dataset. Approximately 88.7% of the 10,000 MW periods contain positive Scottish downward wind BOA activity, with mean downward BOA of approximately 2,658 MW.

By comparison, the remaining observations have an event rate of approximately 60.4% and mean downward BOA of approximately 1,117 MW.

The 10,000 MW state is therefore potentially influential. However, there is insufficient evidence to classify these observations as erroneous or to interpret the published value as meaning that the B6 constraint was disabled.

For this reason, the observations are retained in the main analysis and explicitly flagged. Later robustness tests examine whether the predictive results remain when the 10,000 MW state is excluded.

8 Methodology

8.1 Modelling Sample and Chronological Evaluation

The predictive analysis uses the verified-wind sample of 31,532 settlement periods. Rather than randomly mixing observations across time, the data are divided chronologically.

The earlier modelling sample contains 25,198 settlement periods across 525 days from 22 April 2024 to 6 October 2025. The later evaluation period contains 6,334 settlement periods across 132 days from 7 October 2025 to 23 April 2026.

The event rate is approximately 59% in the earlier sample and 69% in the later period. System conditions are therefore not completely stable through time. A chronological split is preferred because it provides a more realistic test of whether relationships estimated from earlier observations remain useful under later conditions.

The later period was inspected during the development of the analysis. It is therefore described as a later evaluation period rather than a completely untouched confirmatory test set. Internal chronological validation is used where model specification choices are required.

8.2 Event Model

The first modelling stage predicts whether any Scottish downward wind BOA activity occurs during settlement period t .

Define

$$Y_t = \begin{cases} 1, & C_t > 0, \\ 0, & C_t = 0, \end{cases} \quad (6)$$

where C_t is the aggregated Scottish downward wind BOA measure.

Two logistic regression specifications are compared.

The controls-only model is

$$M_0 : P(Y_t = 1) = f(W_t, D_t, \text{calendar}_t), \quad (7)$$

where W_t is the day-ahead wind forecast and D_t is the day-ahead demand forecast.

The B6-enhanced model adds day-ahead B6 headroom:

$$M_1 : P(Y_t = 1) = f(W_t, D_t, H_t^{B6}, \text{calendar}_t). \quad (8)$$

Calendar controls consist of month, day of week and settlement period. Continuous variables are standardised before estimation, while the calendar variables are represented using one-hot encoding.

Comparing M_0 and M_1 provides a direct test of whether B6 headroom adds predictive information beyond the general system conditions already represented by forecast wind and demand.

8.3 Probability Calibration

The event probability is later combined with a conditional magnitude forecast. For this reason, probability calibration matters in addition to ranking performance.

The earlier modelling sample is divided chronologically once more. Observations from 22 April 2024 to 19 June 2025 are used to estimate the event model, while observations from 20 June 2025 to 6 October 2025 form a separate calibration block.

Platt scaling is applied to the raw logistic model score. A second logistic regression is estimated on the calibration period to map the original model score to an observed event probability.

This calibration stage remains separate from both the underlying model-fit period and the later evaluation period.

Calibration is assessed using the Brier score and log loss. Ranking measures such as ROC AUC and average precision are also reported, although these are not expected to change materially under a monotonic probability calibration.

For the final two-part comparison, the same fit/calibration dates and Platt-scaling procedure are applied independently to both the controls-only event model M_0 and the B6-enhanced event model M_1 . This keeps the probability treatment consistent across the two forecasting systems.

8.4 Positive-Event Magnitude Model

The second modelling stage estimates the size of Scottish downward wind BOA activity conditional on positive activity occurring.

The conditional outcome is therefore

$$C_t \mid C_t > 0. \quad (9)$$

This outcome is strictly positive and right-skewed, so Gamma regression with a log link is used.

The controls-only magnitude model contains forecast wind, forecast demand and the same calendar variables used in the event model.

A second specification adds B6 headroom as a single linear predictor. This provides an initial test of whether continuous headroom improves prediction of positive-event magnitude.

8.5 Nonlinear B6 Specification

Exploratory analysis suggests that the relationship between B6 headroom and downward BOA magnitude is not well represented by a single linear term.

A third magnitude specification is therefore considered:

$$G_2 : E(C_t \mid C_t > 0) = g(W_t, D_t, s(H_t^{B6}), I_t^{10GW}, \text{calendar}_t), \quad (10)$$

where $s(H_t^{B6})$ is a spline transformation of B6 headroom and I_t^{10GW} is an indicator for settlement periods in which the published B6 limit equals 10,000 MW.

The spline uses five quantile-based knots and a cubic basis. The Gamma regression is estimated with a regularisation parameter of 0.1.

Model selection is carried out using an internal chronological validation block within the earlier modelling sample rather than by choosing the specification that performs best in the later evaluation period.

Three candidate magnitude models are compared:

- G_0 : forecast wind, forecast demand and calendar controls;
- G_1 : G_0 plus linear B6 headroom;
- G_2 : G_0 plus nonlinear B6 headroom and the 10,000 MW indicator.

After the preferred specification is selected using the internal validation period, it is re-estimated using the full earlier positive-event sample before being applied to the later evaluation period.

8.6 Combined Expected Downward BOA

The event and magnitude stages are combined to obtain an unconditional settlement-period forecast.

For predictor information X_t ,

$$E(C_t | X_t) = P(C_t > 0 | X_t) \times E(C_t | C_t > 0, X_t). \quad (11)$$

The first component represents the calibrated probability of downward wind BOA activity occurring. The second represents the predicted magnitude conditional on an event.

This structure is useful because zero and positive outcomes are generated as separate modelling problems. A settlement period can therefore receive a high expected BOA forecast either because an event is very likely, because its conditional magnitude is expected to be large, or because both conditions apply.

Two complete forecasting systems are compared. The controls-only system combines the Platt-calibrated M_0 event probabilities with the controls-only Gamma magnitude model. The B6-enhanced system combines the Platt-calibrated M_1 event probabilities with the selected nonlinear B6 magnitude model. Both event models use the same chronological model-fit and calibration blocks.

The difference in performance therefore compares a controls-only forecasting system with a B6-enhanced system under the same probability-calibration procedure. The B6-enhanced system differs through the inclusion of B6 information in both the event stage and the selected nonlinear magnitude stage.

8.7 Forecast Evaluation Metrics

Event occurrence is evaluated using ROC AUC, average precision, Brier score and log loss.

ROC AUC measures the ability of the model to rank event periods above non-event periods across possible classification thresholds. Average precision focuses on ranking performance for the positive class. The Brier score and log loss evaluate the quality of the predicted probabilities, with lower values indicating better probabilistic forecasts.

Positive-event magnitude is evaluated using mean absolute error, root mean squared error, R^2 and mean Gamma deviance.

The combined expected-volume forecast is evaluated using

$$MAE = \frac{1}{n} \sum_{t=1}^n |C_t - \hat{C}_t|, \quad (12)$$

and

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (C_t - \hat{C}_t)^2}. \quad (13)$$

Out-of-sample R^2 is also reported as a summary measure of predictive fit over the later evaluation period.

9 Results

9.1 Descriptive Relationship Between B6 Headroom and BOA

The exploratory results show a strong inverse relationship between day-ahead B6 headroom and subsequent Scottish downward wind BOA activity.

Event frequency is approximately 99.8% in the lowest B6 headroom decile and generally declines as headroom increases, reaching approximately 20.3% in the ninth decile. Mean downward BOA follows a similar pattern, falling from approximately 3,638 MW in the lowest-headroom decile to approximately 189 MW in the ninth.

This relationship alone does not establish that B6 conditions are responsible for the balancing actions. Forecast wind is associated with both tighter B6 headroom and greater downward wind activity. The predictive models therefore test whether B6 retains information after these broader system conditions are taken into account.

9.2 Predicting the Occurrence of Downward Wind BOA

Table 2 compares the controls-only event model with the specification that adds continuous B6 headroom.

Table 2: Later-evaluation performance of the event models				
Model	ROC AUC	Avg. precision	Brier score	Log loss
Controls only	0.817	0.909	0.172	0.530
Controls + B6 headroom	0.862	0.936	0.149	0.475

Adding B6 headroom improves performance across all four measures. ROC AUC rises from 0.817 to 0.862 and average precision increases from 0.909 to 0.936. At the same time, the Brier score falls from 0.172 to 0.149 and log loss falls from 0.530 to 0.475.

The improvement is important because forecast wind, forecast demand and calendar conditions are already included in the controls-only model. B6 headroom therefore contains incremental information about the occurrence of downward wind BOA beyond those variables.

The strongest contribution of B6 in the project is found at this event-occurrence stage.

9.3 Probability Calibration

Because event probabilities are subsequently multiplied by the conditional magnitude forecast, the absolute level of the predicted probability also matters.

The B6 event model is therefore re-estimated using the earlier model-fit period and calibrated on the separate chronological calibration block.

Before calibration, this specification produces a Brier score of approximately 0.150 and log loss of 0.475 in the later evaluation period. Platt scaling reduces these to approximately 0.142 and 0.436 respectively.

ROC AUC and average precision remain approximately unchanged at 0.863 and 0.937 because calibration changes the probability scale without materially changing the ordering of observations.

The calibrated probabilities are used in the final two-part B6 forecast.

9.4 Predicting Positive-Event Magnitude

The contribution of B6 is weaker when the task changes from identifying whether an event occurs to predicting its magnitude conditional on occurrence.

A linear B6 specification does not improve the positive-event magnitude model. In the initial comparison, adding B6 headroom as a single linear term increases MAE and RMSE relative to the controls-only Gamma model.

Internal chronological validation instead supports a nonlinear specification using a spline transformation of B6 headroom together with an indicator for the 10,000 MW B6-limit state.

After model selection and re-estimation on the full earlier positive-event sample, the nonlinear B6 model produces the later-evaluation results shown in Table 3.

Table 3: Positive-event magnitude performance

Model	MAE (MW)	RMSE (MW)	R^2	Gamma deviance
Controls only	1010.7	1286.8	0.350	0.828
Nonlinear B6	996.5	1282.3	0.354	0.799

The improvement is modest. MAE falls by approximately 14 MW, while the changes in RMSE and R^2 are small. Mean Gamma deviance also improves from approximately 0.828 to 0.799.

These results suggest that B6 contains some additional information about the size of positive balancing actions, but substantially less than it contains about whether an event occurs at all. The failure of the linear specification also indicates that the relationship between headroom and positive-event magnitude is not well described by a single linear effect.

9.5 Combined Expected Downward BOA

The final comparison combines the calibrated event probability with the conditional magnitude forecast to produce an expected downward BOA value for every settlement period.

Table 4 compares the complete controls-only and B6-enhanced systems.

Table 4: Combined expected downward BOA performance

Model	MAE (MW)	RMSE (MW)	R^2
Controls only	899.8	1230.8	0.47
B6-enhanced	860.1	1192.9	0.50

With the same chronological probability-calibration procedure applied to both event models, the B6-enhanced system records an MAE approximately 40 MW lower than the controls-only system, a difference of 4.4%. RMSE is approximately 38 MW lower, while R^2 rises from approximately 0.47 to 0.50.

The complete B6-enhanced system therefore improves the later-period settlement-level forecast. The gain is modest relative to the predictive power already provided by wind, demand and calendar conditions, which is consistent with B6 adding incremental rather than dominant information.

There is also evidence of level bias in the later period. Mean observed downward BOA is approximately 1,549 MW, compared with a mean B6-enhanced forecast of approximately 1,780 MW. The model therefore overpredicts the average level by around 231 MW, or approximately 15%.

This level error is important because improved discrimination does not imply perfect calibration to a later operating regime. The model captures meaningful cross-sectional and temporal variation while still exhibiting systematic forecast bias.

9.6 Interpretation of Incremental B6 Information

Taken together, the results distinguish between three different claims.

First, general system conditions already contain substantial predictive information. Forecast wind, forecast demand and calendar variables produce a reasonably strong controls-only model, so B6 should not be treated as the dominant explanation for Scottish downward wind BOA activity.

Second, B6 adds information beyond those controls. The clearest improvement occurs in predicting whether downward activity takes place. This is consistent with B6 headroom acting as an informative day-ahead measure of transmission conditions that is not completely captured by forecast wind and demand alone.

Third, the contribution of B6 to predicting the exact magnitude of balancing activity is smaller and nonlinear. This is plausible in a system where the final volume of balancing action may depend on several interacting constraints, network conditions and operational decisions that are not represented by a single boundary measure.

The results therefore support a predictive interpretation rather than a causal one. They show that published day-ahead B6 information improves forecasts of subsequent Scottish downward wind BOA activity, but they do not establish that the observed actions were caused specifically by B6.

10 Robustness

The main results rely on several modelling and data-construction choices. This section tests whether the incremental predictive value associated with B6 remains under alternative treatments of the unusual 10,000 MW limit state, an alternative definition of B6 stress and different parts of the evaluation period.

10.1 Evaluation Excluding the 10,000 MW B6 Limit State

The first robustness test removes settlement periods with a published B6 limit of exactly 10,000 MW from the later evaluation sample while leaving the fitted models unchanged.

There are 46 such observations in the later evaluation period, leaving 6,288 settlement periods for comparison.

The controls-only system records an MAE of approximately 900 MW and an RMSE of 1,232 MW. The B6-enhanced system records an MAE of approximately 860 MW and an RMSE of 1,192 MW.

Out-of-sample R^2 is approximately 0.46 for the controls-only model and 0.50 for the B6-enhanced model.

The improvement associated with B6 therefore remains when the unusual 10,000 MW observations are removed from evaluation. The main result is not being generated solely by unusually large published B6 limits.

10.2 Full Re-estimation Excluding the 10,000 MW B6 Limit State

A stricter robustness test removes the 10,000 MW observations throughout the modelling process rather than only at evaluation.

These observations are excluded from event-model fitting, probability calibration, magnitude-model estimation and the later evaluation sample. The complete modelling system is then re-estimated without them.

Table 5 reports the resulting end-to-end forecast performance.

Table 5: Forecast performance after full exclusion of the 10,000 MW B6 state

Model	MAE (MW)	RMSE (MW)	R^2
Controls only	892.9	1223.6	0.47
B6-enhanced	840.3	1172.8	0.51

The B6-enhanced model again performs better than the controls-only system. The reduction in MAE is approximately 53 MW and the reduction in RMSE is approximately 51 MW.

The incremental improvement is therefore preserved, and slightly larger, when the unusual B6 state is removed completely from model development and evaluation.

10.3 Alternative Definition of B6 Stress

The main specification represents B6 conditions using continuous headroom measured in MW. This retains information about the degree to which forecast flow approaches or exceeds the published transfer limit.

As a robustness check, B6 stress is instead represented by a simple binary indicator equal to one when headroom is negative:

$$I_t^{neg} = \begin{cases} 1, & H_t^{B6} < 0, \\ 0, & H_t^{B6} \geq 0. \end{cases} \quad (14)$$

Table 6 compares the event models.

Table 6: Event-model performance under alternative B6 definitions

Model	ROC AUC	Avg. precision	Brier score
Controls only	0.817	0.909	0.172
Binary negative-headroom indicator	0.854	0.933	0.154
Continuous B6 headroom	0.862	0.936	0.149

The binary indicator performs materially better than the controls-only specification, showing that even a coarse distinction between negative and non-negative headroom contains useful predictive information.

Continuous headroom performs better again. This suggests that the result is not dependent on one precise functional representation of B6 stress, while also indicating that information about the degree of available headroom is useful beyond simply observing whether the published limit has been crossed.

10.4 Temporal Stability

The final robustness check examines end-to-end forecast error separately by calendar month within the later evaluation period.

The B6-enhanced model records lower MAE than the controls-only system in six of the seven represented months. The largest relative improvements occur in October and December 2025, while performance in January 2026 is approximately neutral, with B6 MAE around 0.5% higher than the controls-only model.

The improvement is therefore not confined to a single month. However, its size varies through time.

October and November contain relatively few verified-wind observations, so estimates for these months should be interpreted more cautiously than those based on longer periods.

Taken together, the robustness tests support the same overall conclusion as the main analysis. B6 information continues to improve prediction after the unusual 10,000 MW state is removed, when B6 stress is represented more coarsely and across most months of the later evaluation period.

11 Limitations

The results should be interpreted within several important limitations.

First, the analysis is predictive rather than causal. A relationship between low B6 headroom and subsequent Scottish downward wind BOA activity does not establish that B6 caused those actions. The Scottish transmission system contains several interconnected and nested boundaries, and individual balancing actions may reflect multiple simultaneous network and system requirements. The analysis therefore tests whether B6 is informative about later balancing activity, not whether each action can be attributed directly to B6.

Second, forecast wind is an important control but does not provide a complete representation of the geographical distribution of Scottish generation. The modelling sample uses the verified day-ahead wind forecast available in the project dataset. A more geographically detailed forecast, particularly one separating Scottish generation from the rest of Great Britain, could better capture the generation directly exposed to Scottish transmission constraints.

Third, the outcome is constructed from accepted downward Balancing Mechanism actions for identified Scottish wind units. This provides an observable and reproducible measure of balancing activity, but it is not a complete engineering measure of physical wind curtailment. The data do not provide a direct causal label identifying which transmission constraint motivated each action.

Fourth, B6 headroom is based on published day-ahead forecast flows and transfer limits rather than realised power flows. This is intentional because the project asks what

could have been known before delivery, but it means the variable represents expected transmission conditions rather than the final physical state of the network.

Fifth, the modelling period is limited by the availability of consistent input data. The primary B6 analysis begins on 22 April 2024 because of the identified change in the published day-ahead flow methodology. The verified-wind modelling sample ends on 23 April 2026 and contains 657 days. The results therefore describe a relatively recent operating period rather than a long historical sample covering many different market and network regimes.

Sixth, the later evaluation period was inspected during development of the analysis. It therefore should not be interpreted as a completely untouched confirmatory test set. Internal chronological validation was used for model selection where possible, particularly for the nonlinear magnitude specification, but a genuinely prospective evaluation on future observations would provide a stronger test of model stability.

Seventh, settlement periods are not statistically independent in the same way as randomly sampled observations. Wind conditions, demand, transmission states and balancing activity can persist across neighbouring periods. The chronological design reduces the risk of information leaking from future observations into earlier model fitting, but it does not remove serial dependence from the underlying electricity system.

Finally, the models deliberately use a relatively small set of day-ahead variables. They do not attempt to reconstruct NESO’s full transmission model or incorporate every potentially relevant driver, such as detailed generation location, interconnector conditions, outages across neighbouring boundaries or the full set of operational constraints. This limits forecast accuracy, but also makes the incremental contribution of B6 easier to interpret.

These limitations mean that the results should not be read as a complete model of Scottish constraint management. They provide evidence that published B6 information contains useful day-ahead predictive information within the analysed sample, while leaving substantial scope for additional network and market information to improve the forecast.

12 Conclusion

This project examined whether published day-ahead conditions on the B6 transmission boundary contain useful information about subsequent Scottish downward wind Balancing Mechanism activity.

The descriptive analysis shows a clear relationship between B6 headroom and later balancing activity. Settlement periods with low headroom are associated with substantially higher event frequencies and larger downward BOA volumes. Forecast wind explains part of this relationship because high-wind periods are associated with both tighter B6 conditions and greater downward wind activity. However, the relationship with B6 does

not disappear once these broader system conditions are considered.

The predictive results provide the clearest evidence of this incremental information. Adding B6 headroom to a model containing forecast wind, forecast demand and calendar controls raises event-model ROC AUC from 0.817 to 0.862 and improves the associated probability forecasts. B6 contributes less to predicting the exact magnitude of positive events, where the additional relationship is smaller and appears to be nonlinear.

In the end-to-end implementation, both event models receive the same chronological Platt-calibration treatment. The B6-enhanced system records lower settlement-period error than the controls-only system, with MAE of approximately 860 MW compared with 900 MW and R^2 increasing from approximately 0.47 to 0.50. The largest incremental contribution still comes from the event stage, where B6 materially improves identification of whether downward wind action occurs.

The robustness analysis supports the same conclusion. The improvement remains when the unusual 10,000 MW B6-limit observations are removed, when B6 stress is represented using only a negative-headroom indicator, and across most months in the later evaluation period.

The main conclusion is therefore not that B6 alone determines Scottish wind balancing activity. Rather, published B6 headroom contains useful day-ahead information about the probability of subsequent downward wind action, even after other observable system conditions are taken into account. This demonstrates how publicly available transmission information can be combined with market and forecast data to improve understanding of short-term physical risk in the GB electricity system.

A Supplementary Distributions

The following distributions provide additional context for the modelling choices used in the main analysis.

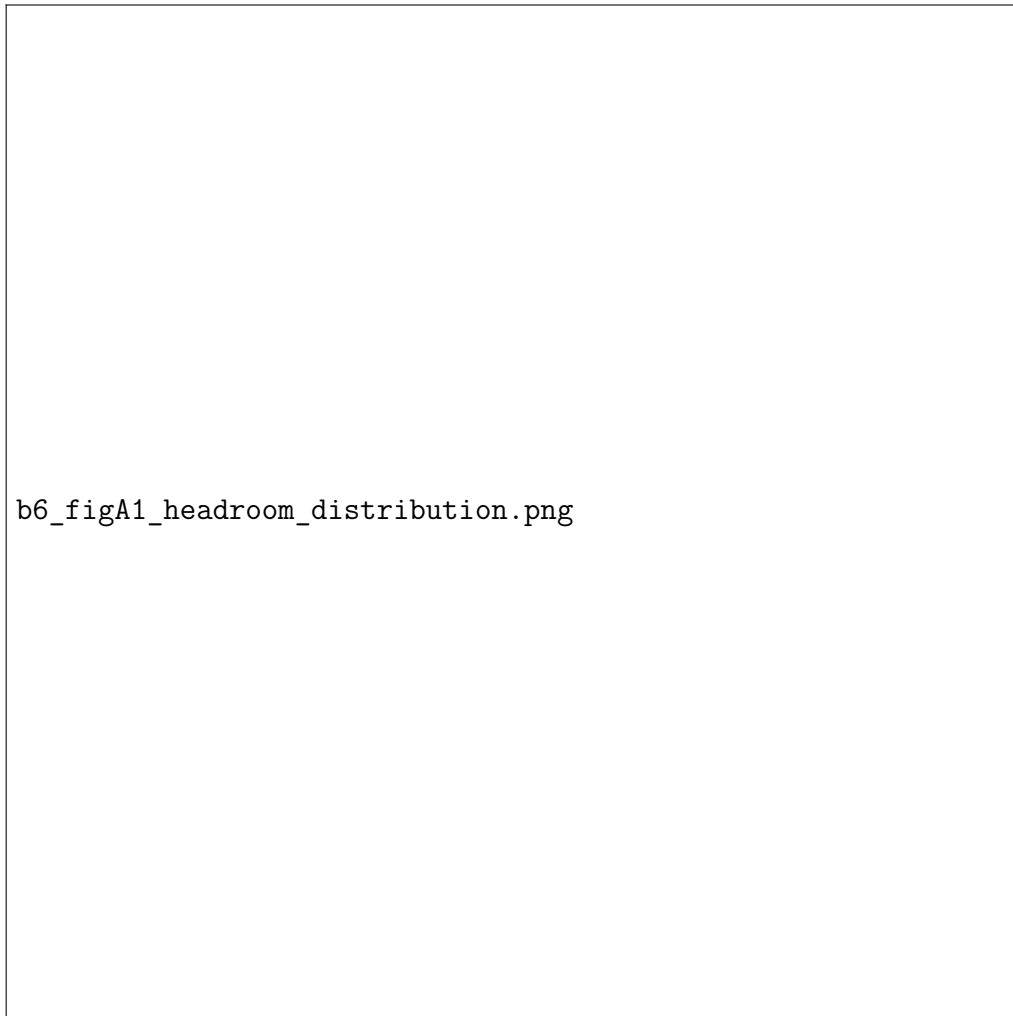


Figure 5: Distribution of day-ahead B6 headroom in the post-methodology-change analytical sample. The distribution spans both negative and positive values and contains a visible high-headroom tail.



Figure 6: Distribution of positive Scottish downward wind BOA volumes. The positive-event distribution is right-skewed, which supports modelling event occurrence separately from positive-event magnitude and motivates the Gamma specification used for the conditional magnitude stage.

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